

7.7 CV-CIM: A 28nm XOR-Derived Similarity-Aware Computation-in-Memory for Cost-Volume Construction

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Cost-volume construction, which accurately computes the similarities between pixels in paired images, is a fundamental kernel of stereo vision processing and has been directly used in robotic, autopilot, and AR/VR applications. However, the large parameter size and consecutive data accesses of real-time cost-volume construction (>30fps) exerts a high demand on the memory bandwidth (0.254Tb/s) and operation (391GOPs). A promising candidate to resolve the memory bottleneck is computation-in-memory (CIM), which provides computing parallelism based on a structured array [1-7]. However, to deploy CIM applications one needs to overcome three challenges: 1) Considering various image properties and frame rate scenarios, cost functions are pattern-dependent and could be, in essence, classified as distances between pixels. CIM only supports restricted-vector Boolean logic, but distance operations require combining variable pixel/vector arithmetic functions. 2) Cost functions are sensitive to the disturbance caused by the reduced CIM computing margin, which is even worse with higher output. Furthermore, distance computing cannot take advantage of input sparsity to reduce the output. 3) The sliding search plane alternates activating weights and incurs data loading overheads, which mismatches the rigid structured array and leads to low utilization.

To resolve these challenges, this paper presents a CV-CIM with the following features: 1) XOR-derived logic integrates arithmetic functions and hybrid vector/pixel XOR-CIMs to compute distance metrics. 2) A similar-aware XOR sparse transform enlarges bitwise sparsity by discovering pixel similarities. A sparse mode CIM captures the sparsity and reduces the output, which improves the sense margin. 3) Dual degrees of freedom peripherals collectively improve the utilization. Row-wise peripherals dynamically skip row computing and support shifted regions. Column-wise freedom unfreezes idle columns to hide loading latency.

Figure 7.7.2 presents the overall architecture of the CV-CIM, including a hybrid XOR-CIM core, a top-level controller, a digital merge unit (DMU), a BIST unit, and other peripherals. The main component contains 4 hybrid XOR groups, each supporting reconfigurable cost volume functions with collaborative 1 pixel-CIM and 8 vector-CIMs. Each CIM array is configured with a 16x64 cell, and 32 XOR-derived units that could activate simultaneously. The pixel-CIM uses an XOR gate and 4 area-optimized full adders to form a ripple-carry adder that merges two adjacent 4b values. The results generated by the pixel-CIM are shared by 8 vector-CIMs, which perform current-based XOR computing and are quantized by a similarity-aware ADC. The DMU processes CIM results based on their bit significance or similarity depending on the CIM mode. A canary-style BIST is embedded to evaluate the accuracy of the vector-CIM under PVT variation.

Figure 7.7.3 illustrates the functionality of the XOR-based distance compute. The XOR acts as a basic logic to accommodate all distance tasks due to its similar nature and transformable property. Considering the pos/negative value transform, this work adopts a multi-bit binary representation (MBR), where each bit equals ± 1 and maintains bit significance. The principal functions, add, compare, and MAC, are developed based on the XOR property. Eventually, a combination of pixel/vector-wise functions realizes distance computing. A hybrid CIM design, detailed in Fig. 7.7.3, accomplishes the entire computing flow. The pixel XOR-CIM utilizes read-out data to perform rail-to-rail logic and to gather neighboring results. To reduce the full-adder (FA) size, a pre-computation FA is configured. The optimization functions by utilizing the pre-processed IN/W and digital XOR to finish partial logic ($R0 \wedge R1$) of SUM/COUT. Pixel-CIM is specific for element processing and the results separately act as inputs to the Vector-CIM, which avoids multi-stage accumulator. To efficiently perform vector computing and accumulation, Vector-CIM employs a current-based method. Input controls the gate of N0/N2/P1 and weight in the cell decides the on/off of P0/N1. The state of footer N3 depends on the mode signal. Under the original mode (mode = 1), the current path is conducted when IN differs from W. All partial results merge in current and are quantized by similarity-aware ADC. Albeit incurring ADC, the design mitigates the challenge of pitch-matching with chunk adder tree. By integrating storage and distance computing units, CV-CIM improves the area density by 2.3x.

Figure 7.7.4 details the similarity-aware XOR transform computing, which narrows down the XOR accumulation (XAC) value to stay within the safe margin. As binomial distribution indicates, the output range is associative with the possibility (output = 1). Unlike AND logic, higher P (output = 1) of XOR induces a wider output range. But another opportunity is noticed that similar inputs produce close sum value. Based on this phenomenon, blocks of inputs compare with the reference pixel and the similarity generates bit-wise sparsity. This similarity transform is completed within Pixel-CIM and the overhead is amortized by sharing similarities with parallel Vector-CIMs. And All-0

input could be detected and bypassed. Subsequently, XOR-based accumulation is converted to 'AND' between similarity and inverted reference results. The conversion allows reducing P (output = 1) with sparser input. Then Vector-CIM (Mode = 0) performs IN & ~W in sparse mode. Those two steps collectively reduce accumulated output by 4.13x and enlarge sense margin by 15%. According to output distribution, 4b quantization covers >99.3% of outputs, and the application concentrates on small candidates, thus a 4b converter is adopted. The similar-aware ADC also observes the sparsity and periodically shut down the comparators in the high-voltage ladder to save power. DMU converts the MBR result to 2's complement for post-processing.

A global search for matching pixels is an NP-hard puzzle. Fortunately, the geometry rule constrains all candidates within the epipolar line. The plane shifts horizontally and forms a search region, but still produces a large footprint. The dual degree of freedom peripheral is dedicated in Fig. 7.7.5. In column dimension, the loading bandwidth is lower than the parallelism of CIM arrays (CA) times column number. And not all columns are activated for computing due to quantization difficulty. Therefore, two constraints are compensated with each other in column peripheral. Specifically, half columns are selected to write and the others perform computing when WL activates. Therefore, idle columns occupied by loading could be released. To reduce the area overhead, the peripheral inherits the intrinsic circuitry, and compute/load arbiter share the address.

Then row-wise peripheral optimizes sliding behavior. Conventional plane-based search accumulates all results within the plane, which distributes 'key' pixels that are from the epipolar line. Instead, an epipolar-first search is proposed to prioritize critical pixels to compute first. Non-significant pixels could be selectively skipped based on pre-winner and pre-tuned threshold comparison. A conservative skip still reduces 11% operation with less than 2.3% accuracy loss. The other benefit is degenerating search in a one-dimension manner. Horizontal pixels are stored in the same row of arrays (row0 in array0, row0 in array1...) successively. The data could be spread as an open chain, where the activating region slides through. According to the 2-color theorem, two addresses could support all slide steps. With two-dimensional techniques, latency reduces by an average of 36%.

Canary-style error detection is employed to analyze Vector-CIM under PVT variation, replicating the most vulnerable scenario and failing earlier than the normal operation. Because of the large distance from the pulse generator, the farthest column encounters the most severe distortion so the determined weight is stored here. Once the output mismatches with the expected result, the comparison equals 1. All comparison signals are sent to Pixel-CIM and the accumulated result is the wrong CA address. Figure 7.7.6 shows the comparison results of CV-CIM, fabricated in 28nm CMOS. The chip works under 0.6-0.9V with 50-286MHz frequency. The peak energy efficiency is 1158TOPs/W at 0.6V. Compared with previous CIM work, various arithmetic function is integrated based on XOR-derived logic. A hybrid vector/pixel paradigm satisfies the computing flow of distances and optimizes XOR computing to enlarge the sense margin by taking advantage of pixel similarity. The flexible array peripheral improves the CIM utilization. Figure 7.7.7 shows the chip and die photo specification with a 0.387mm² area.

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References:

- [1] X. Si et al., "A 28nm 64Kb 6T SRAM Computing-in-Memory Macro with 8b MAC Operation for AI Edge Chips," *ISSCC*, pp. 246-247, 2020.
- [2] W.-S. Khwa et al., "A 65nm 4Kb algorithm-dependent computing-in-memory SRAM unit-macro with 2.3ns and 55.8TOPS/W fully parallel product-sum operation for binary DNN edge processors," *ISSCC*, pp. 496-498, 2018.
- [3] Z. Jiang et al., "C3SRAM: An In-Memory-Computing SRAM Macro Based on Robust Capacitive Coupling Computing Mechanism," *IEEE JSSC*, vol. 55, no. 7, pp. 1888-1897, July 2020.
- [4] S. Yin, Z. Jiang, J.-S. Seo and M. Seok, "XNOR-SRAM: In-Memory Computing SRAM Macro for Binary/Ternary Deep Neural Networks," *IEEE JSSC*, vol. 55, no. 6, pp. 1733-1743, June 2020.
- [5] H. Kim, T. Yoo, T. T. -H. Kim and B. Kim, "Colonnade: A Reconfigurable SRAM-Based Digital Bit-Serial Compute-In-Memory Macro for Processing Neural Networks," in *IEEE JSSC*, vol. 56, no. 7, pp. 2221-2233, July 2021.
- [6] B. Yan et al., "A 1.041-Mb/mm² 27.38-TOPS/W Signed-INT8 Dynamic-Logic-Based ADC-less SRAM Compute-in-Memory Macro in 28nm with Reconfigurable Bitwise Operation for AI and Embedded Applications," *ISSCC*, pp. 188-190, 2022.
- [7] D. Wang, C. -T. Lin, G. K. Chen, P. Knag, R. K. Krishnamurthy and M. Seok, "DIMC: 2219TOPS/W 2569F2/b Digital In-Memory Computing Macro in 28nm Based on Approximate Arithmetic Hardware," *ISSCC*, pp. 266-268, 2022.

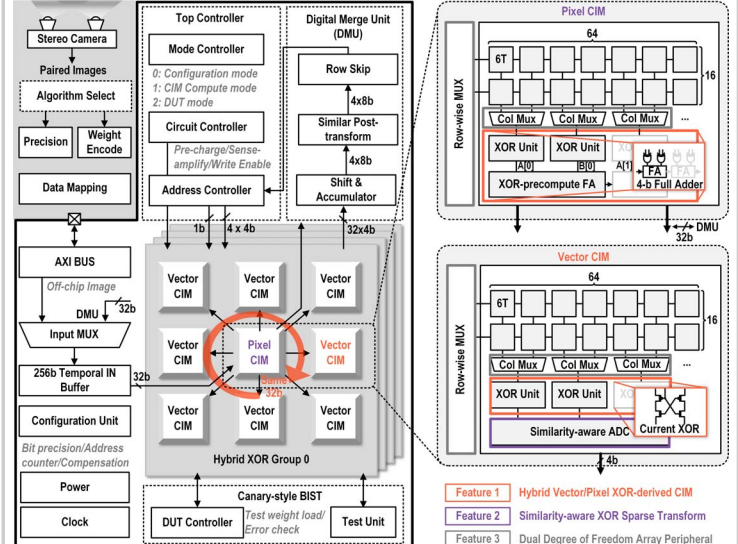


Figure 7.7.2: Overall Architecture of CV-CIM.

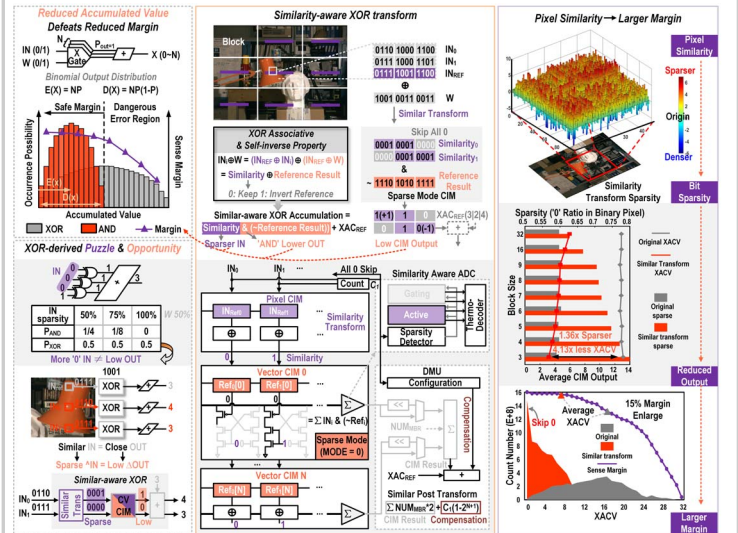


Figure 7.7.4: Similar-aware XOR transform and sparse-mode CIM computing.

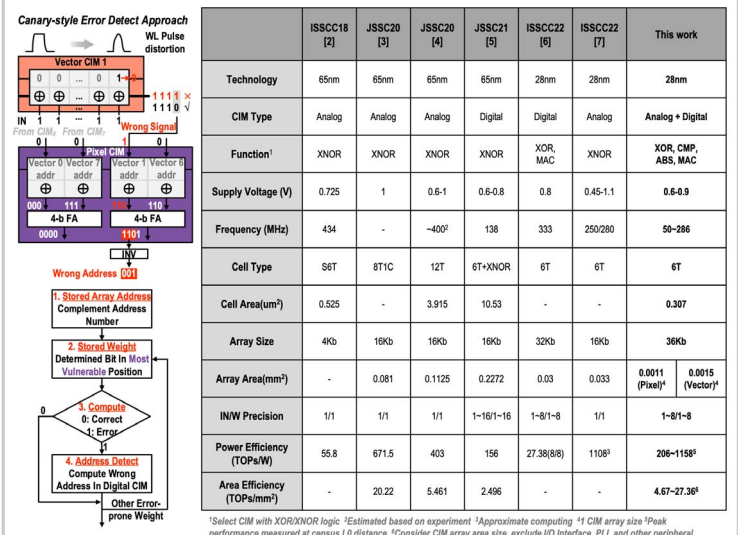


Figure 7.7.6: Canary-style BIST and comparison table to prior CIM accelerators.

